LUIS MANUEL MACÍAS PATIÑO

Data Scientist · Manufacturing & Maintenance

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PROFILE

Data science master's graduate and industrial maintenance engineer with 7 years at an electronics manufacturing plant. I turn plant and finance records into models, web systems and reports used every day: NLP on 51,525 failure records, automated migration of 3,660 bills of materials to SAP, and Flask applications with tests, CI and Docker deployment. Looking for a data scientist or data analyst role in manufacturing, operations or predictive maintenance.

EXPERIENCE

Process Supervisor / Maintenance Supervisor
Idear Electrónica S.A. de C.V. (Grupo BEA), Guadalajara, Mexico · 2020 – present

- Trained a natural-language classifier on 51,525 free-text failure records (139 hand-labeled) that segmented 2,992 failures by replaced component, at 86.8% precision and 88.2% recall.
- Built a bank-transactions web platform for Treasury and Accounting (Flask, MySQL, Active Directory, Alembic, pytest, Jenkins, Docker; 208 commits) with validation, versioning and an audit trail for every upload.
- Designed and coded the Maintenance Management System (Flask, MySQL, Docker, local AI with Ollama/Llama 3.2): work orders, preventive plans, spare-parts inventory, line stops and PDF reports; 184 commits in 51 days.
- Automated, via web scraping, the extraction of 3,660 bills of materials from JetAccess and their conversion to SAP Business One format, in 5 to 6 days of continuous run time.
- Wrote a recursive BOM-explosion algorithm (levels 900 to 100) that computed real consumption and leftovers for 3,093 products over a full production year.
- Publish monthly quality KPIs 6 and 7 (April 2026: 98.87% order closure vs. a 95% target) and automated production, shipping and staffing reports with VBA and Python.
- Moved compressor maintenance from a 6-month schedule to 3,900 hours of actual use with a custom hour meter (hardware and software); wrote the internal automation procedure (ISO 9000) and AS-IS process maps for ISO 27001/37001.

Documentation Assistant · Idear Electrónica S.A. de C.V. · 2019 – 2020

- Production visual aids and product database; work-instruction management.

Electrical Intern · IBM de México · Aug – Dec 2018

- Maintenance of emergency generators and switchboards; maintenance plan for the main substation.

EDUCATION

M.Sc. in Data Science · IEXE Universidad · 2021 – 2023
B.Eng. in Industrial Maintenance · Universidad Tecnológica de Jalisco · 2019 – 2020
Electromechanical Technician · CONALEP Mexicano-Italiano · 2015 – 2019

SKILLS

Data & AI: Python, pandas, NumPy, NLP, machine learning, statistics, ETL, local LLMs (Ollama), Power BI
Engineering: SQL, MySQL, SQLite, Flask, Streamlit, Docker, Git, pytest, Jenkins, LDAP, web scraping, VBA
Plant: SAP PM / SAP Business One, SMT equipment, compressors, electrical, pneumatics, PLC, Arduino, ESP32
Standards: ISO 9001/27001/37001, ANSI ESD S20.20, LOTO OSHA 1910.147, NOM-017-STPS, Lean Six Sigma

CERTIFICATIONS · LANGUAGES

AI (Google · Santander) · SAP · GitHub (Microsoft) · MySQL · Model evaluation with confusion matrices · Lean Six Sigma White Belt · Statistical Process Control · Web development · Advanced Excel
Languages: Spanish (native) · English (in progress, Tecmilenio) · Japanese (intermediate)